

ASSESSMENT TOOL 1 – Coding Challenge

```

* Frame 1: 88 bytes on wire (704 bits)
* Ethernet II
* Internet Protocol Version 4
* User Datagram Protocol
  * Source Port: 59198
  * Destination Port: 53
  * Length: 54
  * Checksum: 0x37cb [correct] (matches partial checksum, not 0x647a, likely caused by "UDP checksum offload")
  * Checksum Status: Good
  * Stream index: 0
  * Stream Packet Number: 1
  * Timestamps
  * UDP payload (46 bytes)
* Domain Name System (query)
```

1. C) Destination Port = 53, well known service – DNS
2. No, UDP does not have sequence numbers or acknowledgement numbers.

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- 3.

Yes, the UDP header fields in DNS and NTP are the **same size: 8 bytes** (Source Port, Destination Port, Length, and Checksum). UDP is suitable for both because it is **fast, connectionless, and has low overhead**, which is ideal for their small data packets.



- **Frame 1: 88 bytes on wire (704 bits)**
- **Ethernet II**
- **Internet Protocol Version 4**
- **User Datagram Protocol**
 - Source Port: 59198
 - Destination Port: 53
 - Length: 54
 - **Checksum: 0x37cb [correct] (matches partner)**
 - Checksum Status: Good
 - Stream index: 0
 - Stream Packet Number: 1
 - **Timestamps**
 - UDP payload (46 bytes)
- **Domain Name System (query)**

- **Frame 1: 90 bytes on wire (720 bits)**
- **Ethernet II**
- **Internet Protocol Version 4**
- **User Datagram Protocol**
 - Source Port: 123
 - Destination Port: 123
 - Length: 56
 - **Checksum: 0x70e1 [correct]**
 - Checksum Status: Good
 - Stream index: 0
 - Stream Packet Number: 1
 - **Timestamps**
 - UDP payload (48 bytes)
- **Network Time Protocol (NTP Version 3)**

4.

▸ Transmission Control Protocol

- Source Port: 21
- Destination Port: 62014
- Stream index: 0
- Stream Packet Number: 1
- Conversation completeness: Incomplete (0)
- TCP Segment Len: 70
- Sequence Number: 0x836f4ffe
- Sequence Number (raw): 2205110270
- Next Sequence Number: 71
- Acknowledgment Number: 0x04846eda
- Acknowledgment number (raw): 75787994
- 1000 = Header Length: 32 bytes (8)
- Flags: 0x018 (PSH, ACK)
- Window: 1448
- Calculated window size: 1448
- Window size scaling factor: -1 (unknown)
- Checksum: 0x9967 [correct]
- Checksum Status: Good
- Urgent Pointer: 0
- Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
- Timestamps
- SEQ/ACK analysis
- TCP payload (70 bytes)

5. The 4-tuple is:

(Source IP address, Source Port, Destination IP address, Destination Port)

For this FTP sample, the ports are **Source Port = 21** and **Destination Port = 62014**.

This 4-tuple uniquely identifies a TCP connection. It helps in **demultiplexing** by allowing the system to determine which TCP connection and application should receive the incoming data.

6. **FTP:** Source Port = 21, Destination Port = 62014

MySQL: Source Port = 56162, Destination Port = 3306

Destination Ports: Different — FTP uses **62014**, while MySQL uses **3306**.

The operating system uses **demultiplexing** to identify the correct application for each packet. It checks the TCP connection information, especially the **destination port number**. Port **3306** identifies MySQL, while port **62014** belongs to the FTP connection shown. This allows the OS to deliver each packet to the correct application/socket.

* Transmission Control Protocol

- * Source Port: 56162
- * Destination Port: 3306
- * Stream index: 0
- * Stream Packet Number: 1
- * Conversation completeness: Incomplete (0)
- * TCP Segment Len: 22
- * Sequence Number: 0xccd8bbb5
- * Sequence Number (raw): 3436755893
- * Next Sequence Number: 23
- * Acknowledgment Number: 0xcd34967b
- * Acknowledgment number (raw): 3442775675
- * 1000 = Header Length: 32 bytes (8)
- * Flags: 0x018 (PSH, ACK)
- * Window: 513
- * Calculated window size: 513
- * Window size scaling factor: -1 (unknown)
- * Checksum: 0x8389 [correct] (matches partial checksum, not 0xfd4e, likely caused by "TCP checksum offload")
- * Checksum Status: Good
- * Urgent Pointer: 0
- * Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
- * Timestamps
- * SEQ/ACK analysis
- * TCP payload (22 bytes)
- * PDU Size: 22

7.

User Datagram Protocol

- Source Port: 59198
- Destination Port: 53
- Length: 54
- Checksum: 0x37cb [correct] (matches p
- Checksum Status: Good
- Stream index: 0
- Stream Packet Number: 1
- Timestamps
- UDP payload (46 bytes)

Transmission Control Protocol

- Source Port: 21
- Destination Port: 62014
- Stream index: 0
- Stream Packet Number: 1
- Conversation completeness: Incomplete (0)
- TCP Segment Len: 70
- Sequence Number: 0x836f4ffe
- Sequence Number (raw): 2205110270
- Next Sequence Number: 71
- Acknowledgment Number: 0x04846eda
- Acknowledgment number (raw): 75787994
- 1000 = Header Length: 32 bytes (8)
- Flags: 0x018 (PSH, ACK)
- Window: 1448
- Calculated window size: 1448
- Window size scaling factor: -1 (unknown)
- Checksum: 0x9967 [correct]
- Checksum Status: Good
- Urgent Pointer: 0
- Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
- Timestamps
- SEQ/ACK analysis
- TCP payload (70 bytes)

TCP has a larger header than UDP. TCP provides reliability through Sequence Number, Acknowledgment Number, ACK flags, and retransmission, while UDP does not provide these features. TCP is connection-oriented because it establishes a connection using the SYN/SYN-ACK/ACK handshake. UDP is connectionless because it sends datagrams without establishing a connection. In the FTP sample, Sequence Number and Acknowledgment Number show reliability, while the Window Size field provides flow control.

8.

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